AI Subsection Yearly Plan 2025-26

OpenMindsClub
Leads: Fatma & Faycal

Goals

1. Build strong AI fundamentals for juniors (theory + practice)
2. Emphasize math-heavy approach with refreshers in linear algebra, probability, and statistics
3. Develop skills in reading and analyzing research papers
4. Support seniors with project-based learning

Structure

Two-Track System (alternating weekly, 2-hour sessions):

- **Week A**: Practical AI (coding, implementation, hands-on)
- **Week B**: Math for AI (linear algebra, probability, statistics, optimization)

Senior Track (parallel):

- Project-based learning with bi-weekly check-ins
- Milestones: proposal (~Jan) → mid-project presentation (~March) → final showcase (~Nov)
- Less mandatory attendance, more autonomy

Timeline Overview (Approximate)

Dec 2025 - Feb 2026: Foundations

- Python & its libraries, EDA
- Linear algebra, probability basics
- First mini-project for juniors

March - May 2026: Core AI & Deep Learning

- Classical ML, neural networks, backpropagation
- Optimization, calculus for ML
- PFE presentations (March)
- Second mini-project for juniors

June 2026: Paper Reading Begins

- Introduction to reading research papers
- Advanced topics based on interest

Summer School (July-Aug): Intensive teaching period

Sept - Nov 2026: Advanced Topics & Showcase

- More paper reading sessions
- Final projects for juniors
- Senior project presentations
- End-of-year showcase

Key Activities

- **Mini-projects**: 2 guided projects for juniors (Feb, May)
- **Final project**: Portfolio-worthy project (May-Nov)
- **Internal challenge**: Mid-year Kaggle-style competition (April)
- **Senior workshops**: Seniors teach juniors about their project topics (March)
- **Paper reading**: Monthly sessions starting June
- **Showcase event**: Final presentations (November)

Note: Detailed monthly plans will be developed and shared with members at the start of each month.